Exercise 6

Problem 1 ("Normal" Inequality)**.** Let $X \sim \mathcal{N}(\mu, \sigma^2)$ normal distributed with mean μ and variance σ^2, what is the probability $P(X \leq X^2)$ in terms of the *standard normal* c.d.f. Φ?

Problem 2 (Density Transformation)**.** Given $X \sim \text{Exp}(\lambda)$ for $\lambda > 0$, what is the density of

$$Y = \frac{2X}{3 + 4X}.$$

Problem 3 (χ^2_1 Quantiles)**.** Assume $X \sim \chi^2_1$, that is X is chi-squared distributed with 1 degree of freedom. Derive an explicit formula for the p-quantiles x_p of X using the quantile function of the normal distribution.

Moreover, use an "intuitive" approach to derive approximate 68% and 95% quantiles based on the fact that the square of a standard normal is χ^2_1 distributed.

Problem 4 (Moment Generating Function)**.** For a random variable X, an interesting function is the *moment generating function* (m.g.f.) defined as

$$M_X(t) = \mathbb{E}\, e^{tX}$$

for t in an neighborhood of 0 (if the expectation exists.) It has the astonishing property that its derivatives "generate" the raw moments of X as

$$\mathbb{E}\, X^k = \frac{d^k}{dt^k} M_X(t)\bigg|_{t=0}.$$

Find the m.g.f. for $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$ and use it to compute the moments $\mathbb{E}\, X^k$ for $k = 0, 1, 2, 3$.

Problem 5 (Daily Loot Boxes)**.** An ominous online game offers a daily loot box containing either a coin game or a cosmetic item. On each of the first six days of a seven-day period, the box contains a coin game with probability 1/7 and a cosmetic item otherwise (independent from day to day). If no coin game has appeared during the first six days, the seventh day is a guaranteed coin game.

If a coin game appears, you play a randomized first round and win with probability 1/6. Winning awards a number of coins distributed as Binomial$(1000, 0.1)$ while losing awards a number distributed as Binomial$(100, 0.1)$.

1. Derive the p.m.f. of the number of games per week,
2. and the p.m.f. of the number of coins for a single coin game.
3. What is the expected number of coins in a week?